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Data Science Is The New Blue Collar Work



Eric Sandosham, Ph.D. · [Follow](#)

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It's ultimately about creating knowledge.



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Background

I had the pleasure of working with some young interns and juniors from a major global system integrator (SI) recently, and was thoroughly impressed with their

adaptability and agility. I was keen to “test” a hypothesis. So I asked these young talents, who were part of the non-IT divisions, if their chances to ascend to the top-rung MD (managing director) positions within the company were better than their IT division colleagues in the offshore delivery centres. Without too much hesitation, they said “Yes”.

Thus, I “confirmed” my hypothesis.

And so I dedicate my 70th article to discussing why IT jobs have become the “blue collar” jobs of the knowledge economy, and the distinct possibility that the vast majority of data scientists are gravitating towards that same direction too.

(I write a weekly series of articles where I call out bad thinking and bad practices in data analytics / data science which you can find [here](#).)

Workforce Pyramid

Don't get all riled up over my seemingly elitist and condescending narrative. Yes, I am aware that the label “blue collar” is associated with work that has a large physical and manual toil component, and that the label does not imply that the work itself requires little skills or education. And that's exactly how I'm using it. IT workers are obviously skilled and educated. But from a workforce pyramid perspective, blue collar jobs are placed on the lower (and broader) rungs.

A few decades ago, a person with computer programming skills would be in high demand and paid well. And they could rise to the C-suite and beyond (e.g. Bill Gates). Given the claim that the role of a data scientist is among the hottest gig in the 21st century, there is expectation from those in the field that they too should rise to C-suite success. I believe the data scientists will be in for a disappointment.

The reality is that the value placed on technical skills, whether they are manual or digital in orientation, will always wane as the gap between demand and supply closes out. The closing of the gap is driven by the (templatised) scaling of education and training, leading to homogeneity. And in an effort to drive down cost and achieve efficiency, offshoring and centralisation of persons with such technical skills simply reinforces the standardisation of practices and the stripping of autonomy and creativity from role-tasks. This inevitably leads to the role getting pushed down the workforce pyramid. “A dime, a dozen” as the saying goes; it doesn't mean there's no demand for the role / skills, but just that it no longer commands a premium.

Creating Knowledge

To understand how this “templatised scaling” affects data scientists, we need to return back to the topic of the knowledge economy. (I’ve written a few articles about the challenges in the knowledge economy: [Why Every Organisation Needs A Chief Librarian](#), [Why AI Won’t Take Your Job](#)). Using the framework of Input → Activity → Output → Outcome → Impact, we see that most data scientists operate in the “activity → output” zone. They don’t manage the inputs, and often, also not the outcomes. These types of roles lack ambiguity and have limited room for creativity. And thus, limited scope in *creating* knowledge.

And this is the underlying truth: leadership will come from jobs that have ambiguity. Successfully navigating ambiguity requires a healthy dose of creativity. Creativity is the essence of problem-solving and adaptability, the output of which is the creation of knowledge. And those who can create knowledge increase their odds of rising to the top.

On the flip side, coding is very much **unambiguous** work, and therefore, have limited scope for creativity. And increasingly, so is data science. With the ubiquitous use of “brute force” techniques like *grid scanning* (i.e. running through all algorithmic permutations in an automated manner), the discipline is becoming templatised. In my 54th article, [I wrote about what it takes to be a Chief Analytics Officer](#). While I approached it from a different argument, the common thread was still that having abilities to manage ambiguity and creativity will more likely propel one towards senior executive positions. This plays out in the broader data practitioner domain where roles that are engaged in *translation* activities (see my previous [article 10](#) and [article 17](#)) have the highest probabilities to rise to the top.

Conclusion

In every economy (agrarian, industrial, information, knowledge), there are those engaging in ambiguous, abstract work that requires creativity, and those engaging in work that’s deterministic and procedural. Just because a job is in high demand (and pays well) doesn’t mean that those in those roles will ascend the corporate ladder. These high-demand jobs typically get squared off through scaling and standardisation, eventually turning them into the blue collar job of the next generation. There is a good possibility that the practice of data science is headed that way.

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Founder & Partner of Red & White Consulting Partners LLP. A passionate and seasoned veteran of business analytics. Former CAO of Citibank APAC.

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Bhaskar Kumar he/him

9 days ago (edited)



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Vishal Saxena

9 days ago



What this article is discounting almost as a complete blind spot is the possibility that data science will shift left. As machines take over the blue collar aspect the humans of DS will drive the engine or be in a position

to. Such articles that are based mostly on opinions are nauseating frankly and inform nothing.

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Scott Biffin

8 days ago



Excellent sharing 

 6  1 reply

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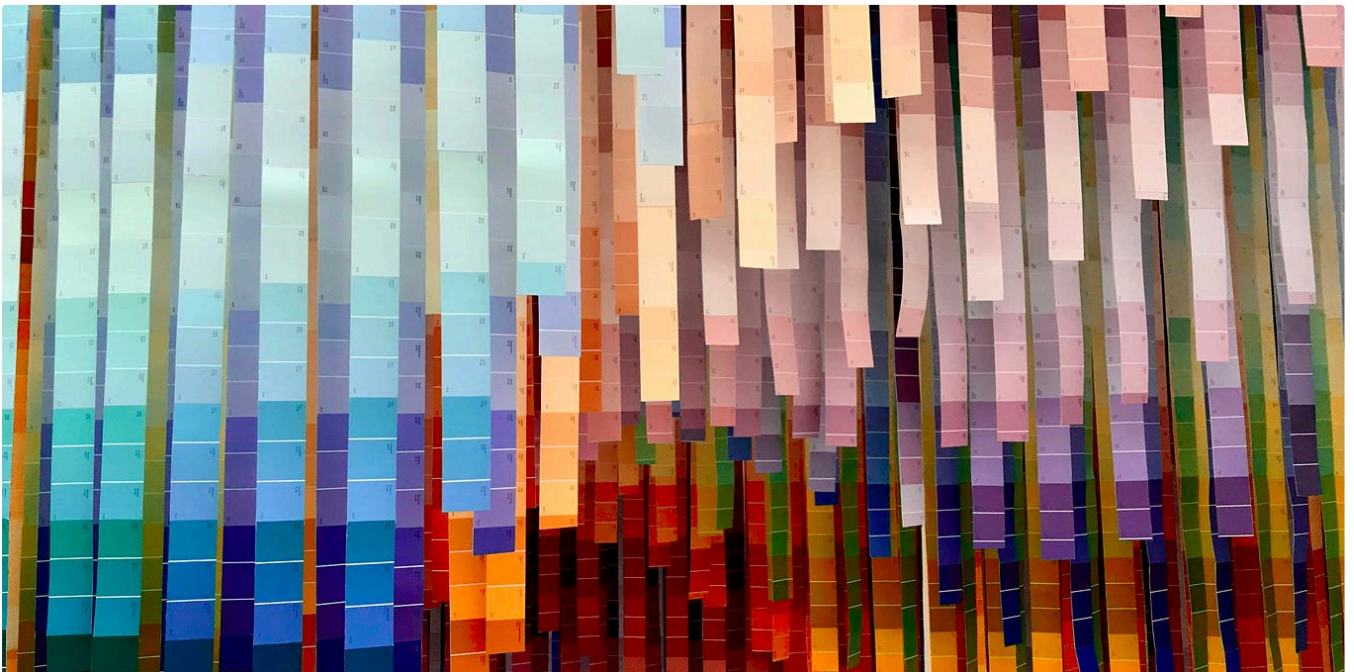


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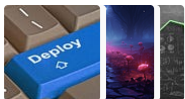
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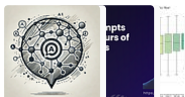
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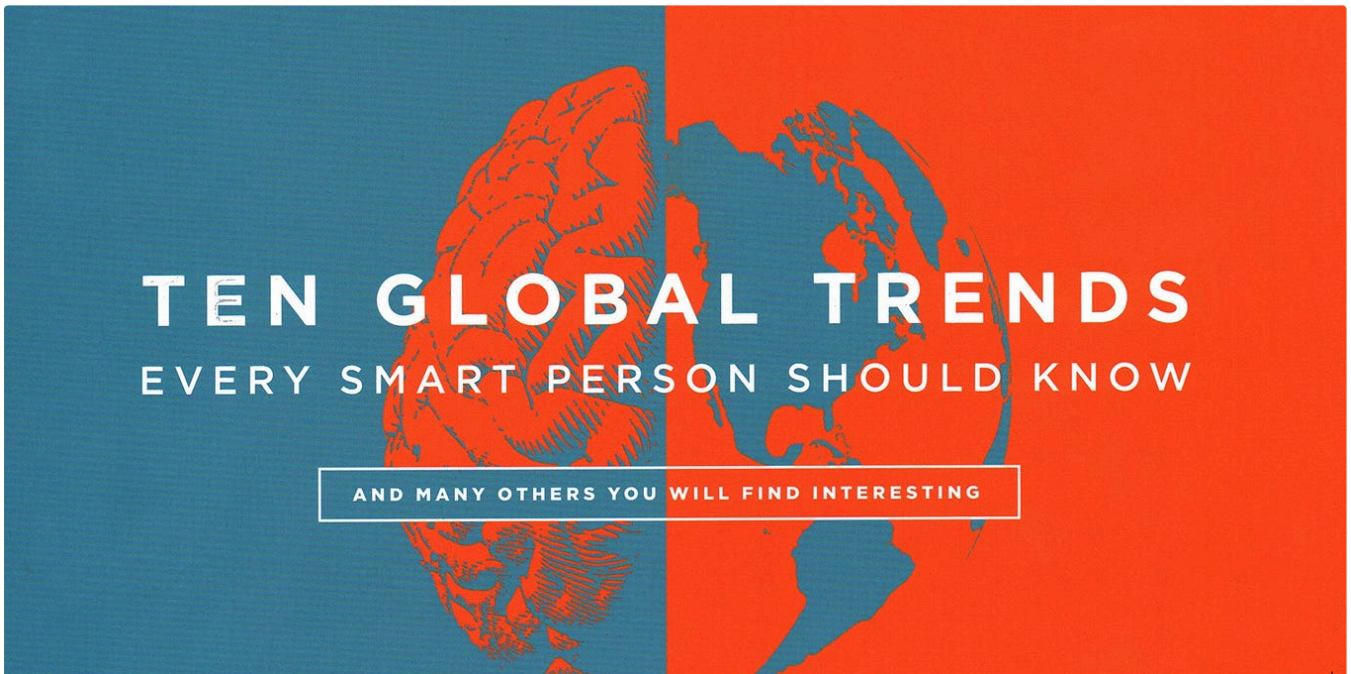
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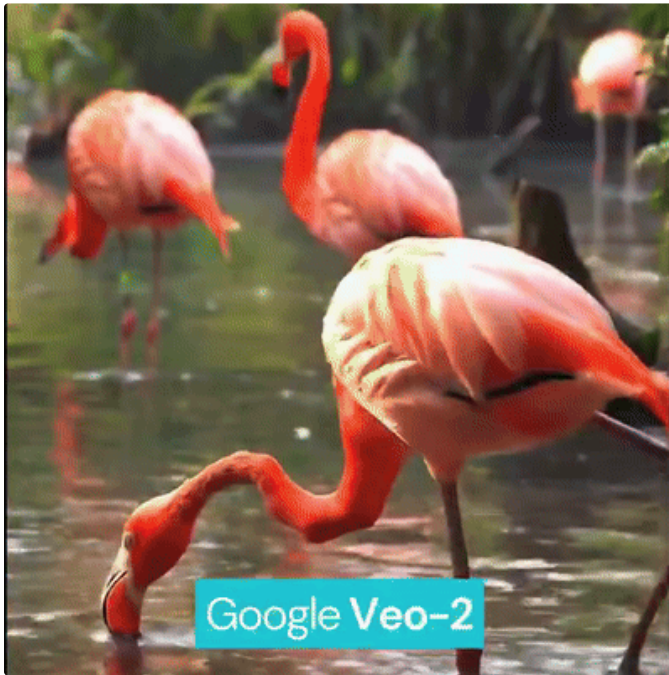


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